EXPERIMENT – 4

DATA MANIPULATION WITH PANDAS

PIN: 241UDS0096

AIM:

To perform data manipulation using the Pandas library, such as selecting specific rows and columns, rounding and updating values, creating calculated columns, conditional filtering, sorting, and renaming columns on the dataset.

Step 1 — Install and Import Pandas Library

In [2]:
```
import pandas as pd
```

Step 2 — Importing the Dataset

In [9]:
```
# Importing the movie dataset
df_241UDS0096 = pd.read_csv("Experiment_4_241UDS0096_Movie_Data.csv")

# Reading the first five rows
df_241UDS0096.head(5)
```

Out[9]:

Columns 1–8 of 13

	Movie_ID	Movie_Title	Genre	Language	Release_Year	Budget_Cr	Tickets_Sold
0	M1001	Pushpa 2: The Rule	Action	Telugu	2024	180	650
1	M1002	Kalki 2898 AD	Sci-Fi	Telugu	2024	600	1050
2	M1003	Devara: Part 1	Action	Telugu	2024	300	520
3	M1004	Hanu-Man	Fantasy	Telugu	2024	40	390
4	M1005	Salaar	Action	Telugu	2023	270	1250

Columns 9–13 of 13 · first column repeated

	Discount_Percent	IMDb_Rating	Available_Seats	Payment_Method	City
0	5.0	4.2	18	UPI	Hyderabad
1	10.0	8.5	35	Card	Vijayawada
2	8.0	6.8	22	UPI	Visakhapatnam
3	12.0	7.8	41	Cash	Pune
4	7.0	6.5	14	Card	Mumbai

Step 3 — Set Index for the DataFrame

In [10]:
```
df_241UDS0096 = df_241UDS0096.set_index("Movie_ID")
df_241UDS0096.head()
```

Out[10]:

Columns 1–8 of 12

Movie_ID	Movie_Title	Genre	Language	Release_Year	Budget_Cr	Tickets_Sold	Discount_Percent
M1001	Pushpa 2: The Rule	Action	Telugu	2024	180	650	5.0
M1002	Kalki 2898 AD	Sci-Fi	Telugu	2024	600	1050	10.0
M1003	Devara: Part 1	Action	Telugu	2024	300	520	8.0
M1004	Hanu-Man	Fantasy	Telugu	2024	40	390	12.0
M1005	Salaar	Action	Telugu	2023	270	1250	7.0

Columns 9–12 of 12 · first column repeated

Movie_ID	IMDb_Rating	Available_Seats	Payment_Method	City
M1001	4.2	18	UPI	Hyderabad
M1002	8.5	35	Card	Vijayawada
M1003	6.8	22	UPI	Visakhapatnam
M1004	7.8	41	Cash	Pune
M1005	6.5	14	Card	Mumbai

Step 4 — Info of the DataFrame

In [11]:
```
df_241UDS0096.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 30 entries, M1001 to M1030
Data columns (total 11 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   Movie_Title       30 non-null     object 
 1   Genre             30 non-null     object 
 2   Language          30 non-null     object 
 3   Release_Year      30 non-null     int64  
 4   Budget_Cr         30 non-null     int64  
 5   Tickets_Sold      30 non-null     int64  
 6   Discount_Percent  30 non-null     float64
 7   IMDb_Rating       30 non-null     float64
 8   Available_Seats   30 non-null     int64  
 9   Payment_Method    30 non-null     object 
 10  City              30 non-null     object 
dtypes: float64(2), int64(4), object(5)
memory usage: 2.8+ KB
```

Step 5 — Columns & Rows Selection: Selecting a Single Column

In [12]:
```
df_241UDS0096["Movie_Title"]
```

Out[12]:

Movie_ID	Movie_Title
M1001	Pushpa 2: The Rule
M1002	Kalki 2898 AD
M1003	Devara: Part 1
M1004	Hanu-Man
M1005	Salaar
M1006	RRR
M1007	Jersey
M1008	Sita Ramam
M1009	Ala Vaikunthapurramuloo
M1010	Arjun Reddy
M1011	Eega
M1012	Mahanati
M1013	Rangasthalam
M1014	Goodachari
M1015	Baahubali 2: The Conclusion
M1016	Magadheera
M1017	Jathi Ratnalu
M1018	Agent Sai Srinivasa Athreya
M1019	C/o Kancharapalem
M1020	Brochevarevarura
M1021	Premalu
M1022	Manjummel Boys
M1023	Aavesham
M1024	2018
M1025	Vikram
M1026	Leo
M1027	Ponniyin Selvan: I
M1028	96
M1029	Soorarai Pottru
M1030	Maharaja

dtype: object

Step 6 — Selecting Multiple Columns

In [13]:
```
df_241UDS0096[["Movie_Title", "Genre", "Language"]].head(3)
```

Out[13]:

Movie_ID	Movie_Title	Genre	Language
M1001	Pushpa 2: The Rule	Action	Telugu
M1002	Kalki 2898 AD	Sci-Fi	Telugu
M1003	Devara: Part 1	Action	Telugu

Step 7 — Select Row Using iloc

In [14]:
```
# Select the second row by position
df_241UDS0096.iloc[1]
```

Out[14]:

	M1002
Movie_Title	Kalki 2898 AD
Genre	Sci-Fi
Language	Telugu
Release_Year	2024
Budget_Cr	600
Tickets_Sold	1050
Discount_Percent	10.0
IMDb_Rating	8.5
Available_Seats	35
Payment_Method	Card
City	Vijayawada

dtype: object

Step 8 — Select Row Using loc

In [15]:
```
# Select a row by its Movie_ID label
df_241UDS0096.loc["M1030"]
```

Out[15]:

	M1030
Movie_Title	Maharaja
Genre	Thriller
Language	Tamil
Release_Year	2024
Budget_Cr	20
Tickets_Sold	1750
Discount_Percent	15.0
IMDb_Rating	8.5
Available_Seats	10
Payment_Method	UPI
City	Hyderabad

dtype: object

Step 9 — Manipulating Data Based on Columns

In [16]:
```
df_241UDS0096.head(2)
```

Out[16]:

Columns 1–8 of 12

Movie_ID	Movie_Title	Genre	Language	Release_Year	Budget_Cr	Tickets_Sold	Discount_Percent
M1001	Pushpa 2: The Rule	Action	Telugu	2024	180	650	5.0
M1002	Kalki 2898 AD	Sci-Fi	Telugu	2024	600	1050	10.0

Columns 9–12 of 12 · first column repeated

Movie_ID	IMDb_Rating	Available_Seats	Payment_Method	City
M1001	4.2	18	UPI	Hyderabad
M1002	8.5	35	Card	Vijayawada

Step 10 — Creating Total Box Office Column

In [17]:
```
# Add a sample ticket price, then estimate gross box-office revenue
df_241UDS0096["Ticket_Price"] = 250
df_241UDS0096["Total_Box_Office_241UDS0096"] = (
    df_241UDS0096["Tickets_Sold"] * df_241UDS0096["Ticket_Price"]
)
df_241UDS0096.head(2)
```

Out[17]:

Columns 1–8 of 14

	Movie_Title	Genre	Language	Release_Year	Budget_Cr	Tickets_Sold	Discount_Percent
Movie_ID							
M1001	Pushpa 2: The Rule	Action	Telugu	2024	180	650	5.0
M1002	Kalki 2898 AD	Sci-Fi	Telugu	2024	600	1050	10.0

Columns 9–14 of 14 · first column repeated

	IMDb_Rating	Available_Seats	Payment_Method	City	Ticket_Price	Total_Box_Office_241UDS0096
Movie_ID						
M1001	4.2	18	UPI	Hyderabad	250	162500
M1002	8.5	35	Card	Vijayawada	250	262500

Step 11 — Creating Discount Amount Column

In [18]:
```
df_241UDS0096["Discount_Amount_241UDS0096"] = (
    df_241UDS0096["Total_Box_Office_241UDS0096"] *
df_241UDS0096["Discount_Percent"] / 100
)
df_241UDS0096.head(2)
```

Out[18]: Columns 1–8 of 15

	Movie_Title	Genre	Language	Release_Year	Budget_Cr	Tickets_Sold	Discount_Percent
Movie_ID							
M1001	Pushpa 2: The Rule	Action	Telugu	2024	180	650	5.0
M1002	Kalki 2898 AD	Sci-Fi	Telugu	2024	600	1050	10.0

Columns 9–15 of 15 · first column repeated

	IMDb_Rating	Available_Seats	Payment_Method	City	Ticket_Price	Total_Box_Office_241UDS0096	Discount_Amount_241UDS0096
Movie_ID							
M1001	4.2	18	UPI	Hyderabad	250	162500	8125.0
M1002	8.5	35	Card	Vijayawada	250	262500	26250.0

Step 12 — Creating Net Revenue Column

In [19]:
```
df_241UDS0096["Net_Revenue_241UDS0096"] = (
    df_241UDS0096["Total_Box_Office_241UDS0096"] -
df_241UDS0096["Discount_Amount_241UDS0096"]
)
df_241UDS0096.head(2)
```

Out[19]:

Columns 1–8 of 16

	Movie_Title	Genre	Language	Release_Year	Budget_Cr	Tickets_Sold	Discount_Percent
Movie_ID							
M1001	Pushpa 2: The Rule	Action	Telugu	2024	180	650	5.0
M1002	Kalki 2898 AD	Sci-Fi	Telugu	2024	600	1050	10.0

Columns 9–15 of 16 · first column repeated

	IMDb_Rating	Available_Seats	Payment_Method	City	Ticket_Price	Total_Box_Office_241UDS0096	Discount_Amount_241UDS0096
Movie_ID							
M1001	4.2	18	UPI	Hyderabad	250	162500	8125.0
M1002	8.5	35	Card	Vijayawada	250	262500	26250.0

Columns 16–16 of 16 · first column repeated

	Net_Revenue_241UDS0096
Movie_ID	
M1001	154375.0
M1002	236250.0

Step 13 — Round Calculated Values

In [20]:
```
df_241UDS0096["Discount_Amount_241UDS0096"] =
df_241UDS0096["Discount_Amount_241UDS0096"].round(2)
df_241UDS0096["Net_Revenue_241UDS0096"] =
df_241UDS0096["Net_Revenue_241UDS0096"].round(2)
df_241UDS0096.head(2)
```

Out[20]: Columns 1–8 of 16

	Movie_Title	Genre	Language	Release_Year	Budget_Cr	Tickets_Sold	Discount_Percent
Movie_ID							
M1001	Pushpa 2: The Rule	Action	Telugu	2024	180	650	5.0
M1002	Kalki 2898 AD	Sci-Fi	Telugu	2024	600	1050	10.0

Columns 9–15 of 16 · first column repeated

	IMDb_Rating	Available_Seats	Payment_Method	City	Ticket_Price	Total_Box_Office_241UDS0096	Discount_Amount_241UDS0096
Movie_ID							
M1001	4.2	18	UPI	Hyderabad	250	162500	8125.0
M1002	8.5	35	Card	Vijayawada	250	262500	26250.0

Columns 16–16 of 16 · first column repeated

	Net_Revenue_241UDS0096
Movie_ID	
M1001	154375.0
M1002	236250.0

Step 14 — Conditional Filtering: Movies with Tickets Sold Greater Than 500

In [21]:
```
df_241UDS0096[df_241UDS0096["Tickets_Sold"] > 500].head(2)
```

Out[21]:

Columns 1–8 of 16

	Movie_Title	Genre	Language	Release_Year	Budget_Cr	Tickets_Sold	Discount_Percent
Movie_ID							
M1001	Pushpa 2: The Rule	Action	Telugu	2024	180	650	5.0
M1002	Kalki 2898 AD	Sci-Fi	Telugu	2024	600	1050	10.0

Columns 9–15 of 16 · first column repeated

	IMDb_Rating	Available_Seats	Payment_Method	City	Ticket_Price	Total_Box_Office_241UDS0096	Discount_Amount_241UDS0096
Movie_ID							
M1001	4.2	18	UPI	Hyderabad	250	162500	8125.0
M1002	8.5	35	Card	Vijayawada	250	262500	26250.0

Columns 16–16 of 16 · first column repeated

	Net_Revenue_241UDS0096
Movie_ID	
M1001	154375.0
M1002	236250.0

Step 15 — High Box Office and Low Available Seats

In [22]:
```
df_241UDS0096[
    (df_241UDS0096["Total_Box_Office_241UDS0096"] > 300000) &
    (df_241UDS0096["Available_Seats"] < 25)
].head(3)
```

Out[22]:

Columns 1–8 of 16

	Movie_Title	Genre	Language	Release_Year	Budget_Cr	Tickets_Sold	Discount_Percent
Movie_ID							
M1005	Salaar	Action	Telugu	2023	270	1250	7.0
M1007	Jersey	Drama	Telugu	2019	25	1850	6.0
M1009	Ala Vaikunthapurramuloo	Drama	Telugu	2020	100	1450	11.0

Columns 9–15 of 16 · first column repeated

	IMDb_Rating	Available_Seats	Payment_Method	City	Ticket_Price	Total_Box_Office_241UDS0096	Discount_Amount_241UDS0096
Movie_ID							
M1005	6.5	14	Card	Mumbai	250	312500	21875.0
M1007	8.1	19	UPI	Bangalore	250	462500	27750.0
M1009	7.3	16	UPI	Hyderabad	250	362500	39875.0

Columns 16–16 of 16 · first column repeated

	Net_Revenue_241UDS0096
Movie_ID	
M1005	290625.0
M1007	434750.0
M1009	322625.0

Step 16 — Movies from Hyderabad OR Mumbai

In [23]:
```
df_241UDS0096[
    (df_241UDS0096["City"] == "Hyderabad") |
    (df_241UDS0096["City"] == "Mumbai")
].head()
```

Out[23]: Columns 1–8 of 16

	Movie_Title	Genre	Language	Release_Year	Budget_Cr	Tickets_Sold	Discount_Percent
Movie_ID							
M1001	Pushpa 2: The Rule	Action	Telugu	2024	180	650	5.0
M1005	Salaar	Action	Telugu	2023	270	1250	7.0
M1009	Ala Vaikunthapurramuloo	Drama	Telugu	2020	100	1450	11.0
M1013	Rangasthalam	Drama	Telugu	2018	60	620	13.0
M1016	Magadheera	Fantasy	Telugu	2009	35	720	7.0

Columns 9–15 of 16 · first column repeated

	IMDb_Rating	Available_Seats	Payment_Method	City	Ticket_Price	Total_Box_Office_241UDS0096	Discount_Amount_241UDS0096
Movie_ID							
M1001	4.2	18	UPI	Hyderabad	250	162500	8125.0
M1005	6.5	14	Card	Mumbai	250	312500	21875.0
M1009	7.3	16	UPI	Hyderabad	250	362500	39875.0
M1013	8.2	31	Card	Mumbai	250	155000	20150.0
M1016	7.7	28	Cash	Hyderabad	250	180000	12600.0

Columns 16–16 of 16 · first column repeated

	Net_Revenue_241UDS0096
Movie_ID	
M1001	154375.0
M1005	290625.0
M1009	322625.0
M1013	134850.0
M1016	167400.0

Step 17 — Movies with IMDb Rating ≥ 8.0 AND Available Seats < 30

In [24]:

```
df_241UDS0096[
    (df_241UDS0096["IMDb_Rating"] ⩾ 8.0) &
    (df_241UDS0096["Available_Seats"] < 30)
]
```

Out[24]:

Columns 1–8 of 16

Movie_ID	Movie_Title	Genre	Language	Release_Year	Budget_Cr	Tickets_Sold	Discount_Percent
M1007	Jersey	Drama	Telugu	2019	25	1850	6.0
M1008	Sita Ramam	Romance	Telugu	2022	30	480	8.0
M1012	Mahanati	Biography	Telugu	2018	25	540	10.0
M1015	Baahubali 2: The Conclusion	Fantasy	Telugu	2017	250	2100	9.0
M1018	Agent Sai Srinivasa Athreya	Mystery	Telugu	2019	2	1150	14.0
M1019	C/o Kancharapalem	Drama	Telugu	2018	1	320	11.0
M1020	Brochevarevarura	Comedy	Telugu	2019	8	760	11.0
M1024	2018	Drama	Malayalam	2023	25	760	11.0
M1030	Maharaja	Thriller	Tamil	2024	20	1750	15.0

Columns 9–15 of 16 · first column repeated

Movie_ID	IMDb_Rating	Available_Seats	Payment_Method	City	Ticket_Price	Total_Box_Office_241UDS0096	Discount_Amount_241UDS0096
M1007	8.1	19	UPI	Bangalore	250	462500	27750.0
M1008	8.6	22	Card	Chennai	250	120000	9600.0
M1012	8.4	21	UPI	Pune	250	135000	13500.0
M1015	8.2	12	Card	Chennai	250	525000	47250.0
M1018	8.3	17	Card	Delhi	250	287500	40250.0
M1019	8.4	26	Cash	Visakhapatnam	250	80000	8800.0
M1020	8.0	24	Cash	Kochi	250	190000	20900.0
M1024	8.0	24	Cash	Delhi	250	190000	20900.0
M1030	8.5	10	UPI	Hyderabad	250	437500	65625.0

Columns 16–16 of 16 · first column repeated

Movie_ID	Net_Revenue_241UDS0096
M1007	434750.0
M1008	110400.0
M1012	121500.0
M1015	477750.0

	Net_Revenue_241UDS0096
Movie_ID	
M1018	247250.0
M1019	71200.0
M1020	169100.0
M1024	169100.0
M1030	371875.0

Step 18 — Filtering Multiple Cities Using .isin()

In [25]:
```
df_241UDS0096[df_241UDS0096["City"].isin(["Hyderabad", "Bangalore", "Pune"])].head()
```

Out[25]: Columns 1–8 of 16

	Movie_Title	Genre	Language	Release_Year	Budget_Cr	Tickets_Sold	Discount_Percent
Movie_ID							
M1001	Pushpa 2: The Rule	Action	Telugu	2024	180	650	5.0
M1004	Hanu-Man	Fantasy	Telugu	2024	40	390	12.0
M1007	Jersey	Drama	Telugu	2019	25	1850	6.0
M1009	Ala Vaikunthapurramuloo	Drama	Telugu	2020	100	1450	11.0
M1012	Mahanati	Biography	Telugu	2018	25	540	10.0

Columns 9–15 of 16 · first column repeated

	IMDb_Rating	Available_Seats	Payment_Method	City	Ticket_Price	Total_Box_Office_241UDS0096	Discount_Amount_241UDS0096
Movie_ID							
M1001	4.2	18	UPI	Hyderabad	250	162500	8125.0
M1004	7.8	41	Cash	Pune	250	97500	11700.0
M1007	8.1	19	UPI	Bangalore	250	462500	27750.0
M1009	7.3	16	UPI	Hyderabad	250	362500	39875.0
M1012	8.4	21	UPI	Pune	250	135000	13500.0

Columns 16–16 of 16 · first column repeated

	Net_Revenue_241UDS0096
Movie_ID	
M1001	154375.0
M1004	85800.0
M1007	434750.0
M1009	322625.0
M1012	121500.0

Step 19 — Filtering Multiple Genres Using .isin()

In [26]:
```
df_241UDS0096[df_241UDS0096["Genre"].isin(["Action", "Drama"])].head()
```

Out[26]:

Columns 1–8 of 16

	Movie_Title	Genre	Language	Release_Year	Budget_Cr	Tickets_Sold	Discount_Percent
Movie_ID							
M1001	Pushpa 2: The Rule	Action	Telugu	2024	180	650	5.0
M1003	Devara: Part 1	Action	Telugu	2024	300	520	8.0
M1005	Salaar	Action	Telugu	2023	270	1250	7.0
M1006	RRR	Action	Telugu	2022	550	150	15.0
M1007	Jersey	Drama	Telugu	2019	25	1850	6.0

Columns 9–15 of 16 · first column repeated

	IMDb_Rating	Available_Seats	Payment_Method	City	Ticket_Price	Total_Box_Office_241UDS0096	Discount_Amount_241UDS0096
Movie_ID							
M1001	4.2	18	UPI	Hyderabad	250	162500	8125.0
M1003	6.8	22	UPI	Visakhapatnam	250	130000	10400.0
M1005	6.5	14	Card	Mumbai	250	312500	21875.0
M1006	7.8	60	Cash	Delhi	250	37500	5625.0
M1007	8.1	19	UPI	Bangalore	250	462500	27750.0

Columns 16–16 of 16 · first column repeated

	Net_Revenue_241UDS0096
Movie_ID	
M1001	154375.0
M1003	119600.0
M1005	290625.0
M1006	31875.0
M1007	434750.0

Step 20 — Filtering Multiple Payment Methods Using .isin()

In [27]:
```
df_241UDS0096[df_241UDS0096["Payment_Method"].isin(["UPI", "Card"])].head()
```

Out[27]:

Columns 1–8 of 16

	Movie_Title	Genre	Language	Release_Year	Budget_Cr	Tickets_Sold	Discount_Percent
Movie_ID							
M1001	Pushpa 2: The Rule	Action	Telugu	2024	180	650	5.0
M1002	Kalki 2898 AD	Sci-Fi	Telugu	2024	600	1050	10.0
M1003	Devara: Part 1	Action	Telugu	2024	300	520	8.0
M1005	Salaar	Action	Telugu	2023	270	1250	7.0
M1007	Jersey	Drama	Telugu	2019	25	1850	6.0

Columns 9–15 of 16 · first column repeated

	IMDb_Rating	Available_Seats	Payment_Method	City	Ticket_Price	Total_Box_Office_241UDS0096	Discount_Amount_241UDS0096
Movie_ID							
M1001	4.2	18	UPI	Hyderabad	250	162500	8125.0
M1002	8.5	35	Card	Vijayawada	250	262500	26250.0
M1003	6.8	22	UPI	Visakhapatnam	250	130000	10400.0
M1005	6.5	14	Card	Mumbai	250	312500	21875.0
M1007	8.1	19	UPI	Bangalore	250	462500	27750.0

Columns 16–16 of 16 · first column repeated

	Net_Revenue_241UDS0096
Movie_ID	
M1001	154375.0
M1002	236250.0
M1003	119600.0
M1005	290625.0
M1007	434750.0

Step 21 — Sorting Table by Tickets Sold (Descending)

In [28]:
```
df_241UDS0096.sort_values("Tickets_Sold", ascending=False).head()
```

Out[28]:

Columns 1–8 of 16

	Movie_Title	Genre	Language	Release_Year	Budget_Cr	Tickets_Sold	Discount_Percent
Movie_ID							
M1015	Baahubali 2: The Conclusion	Fantasy	Telugu	2017	250	2100	9.0
M1007	Jersey	Drama	Telugu	2019	25	1850	6.0
M1030	Maharaja	Thriller	Tamil	2024	20	1750	15.0
M1026	Leo	Action	Tamil	2023	250	1550	13.0
M1009	Ala Vaikunthapurramuloo	Drama	Telugu	2020	100	1450	11.0

Columns 9–15 of 16 · first column repeated

	IMDb_Rating	Available_Seats	Payment_Method	City	Ticket_Price	Total_Box_Office_241UDS0096	Discount_Amount_241UDS0096
Movie_ID							
M1015	8.2	12	Card	Chennai	250	525000	47250.0
M1007	8.1	19	UPI	Bangalore	250	462500	27750.0
M1030	8.5	10	UPI	Hyderabad	250	437500	65625.0
M1026	7.2	13	Card	Mumbai	250	387500	50375.0
M1009	7.3	16	UPI	Hyderabad	250	362500	39875.0

Columns 16–16 of 16 · first column repeated

	Net_Revenue_241UDS0096
Movie_ID	
M1015	477750.0
M1007	434750.0
M1030	371875.0
M1026	337125.0
M1009	322625.0

Step 22 — Sort by Budget (Ascending)

In [29]:
```
df_241UDS0096.sort_values("Budget_Cr", ascending=True).head()
```

Out[29]: Columns 1–8 of 16

	Movie_Title	Genre	Language	Release_Year	Budget_Cr	Tickets_Sold	Discount_Percent
Movie_ID							
M1019	C/o Kancharapalem	Drama	Telugu	2018	1	320	11.0
M1018	Agent Sai Srinivasa Athreya	Mystery	Telugu	2019	2	1150	14.0
M1017	Jathi Ratnalu	Comedy	Telugu	2021	4	130	5.0
M1014	Goodachari	Thriller	Telugu	2018	6	190	6.0
M1020	Brochevarevarura	Comedy	Telugu	2019	8	760	11.0

Columns 9–15 of 16 · first column repeated

	IMDb_Rating	Available_Seats	Payment_Method	City	Ticket_Price	Total_Box_Office_241UDS0096	Discount_Amount_241UDS0096
Movie_ID							
M1019	8.4	26	Cash	Visakhapatnam	250	80000	8800.0
M1018	8.3	17	Card	Delhi	250	287500	40250.0
M1017	7.3	55	UPI	Vijayawada	250	32500	1625.0
M1014	7.7	39	UPI	Bangalore	250	47500	2850.0
M1020	8.0	24	Cash	Kochi	250	190000	20900.0

Columns 16–16 of 16 · first column repeated

	Net_Revenue_241UDS0096
Movie_ID	
M1019	71200.0
M1018	247250.0
M1017	30875.0
M1014	44650.0
M1020	169100.0

Step 23 — Sorting Using Multiple Columns

In [30]:
```
df_241UDS0096.sort_values(["Genre", "Tickets_Sold"], ascending=[True, False]).head()
```

Out[30]:

Columns 1–8 of 16

	Movie_Title	Genre	Language	Release_Year	Budget_Cr	Tickets_Sold	Discount_Percent
Movie_ID							
M1026	Leo	Action	Tamil	2023	250	1550	13.0
M1005	Salaar	Action	Telugu	2023	270	1250	7.0
M1001	Pushpa 2: The Rule	Action	Telugu	2024	180	650	5.0
M1003	Devara: Part 1	Action	Telugu	2024	300	520	8.0
M1023	Aavesham	Action	Malayalam	2024	30	180	8.0

Columns 9–15 of 16 · first column repeated

	IMDb_Rating	Available_Seats	Payment_Method	City	Ticket_Price	Total_Box_Office_241UDS0096	Discount_Amount_241UDS0096
Movie_ID							
M1026	7.2	13	Card	Mumbai	250	387500	50375.0
M1005	6.5	14	Card	Mumbai	250	312500	21875.0
M1001	4.2	18	UPI	Hyderabad	250	162500	8125.0
M1003	6.8	22	UPI	Visakhapatnam	250	130000	10400.0
M1023	7.8	48	UPI	Chennai	250	45000	3600.0

Columns 16–16 of 16 · first column repeated

	Net_Revenue_241UDS0096
Movie_ID	
M1026	337125.0
M1005	290625.0
M1001	154375.0
M1003	119600.0
M1023	41400.0

Step 24 — Updating Data in the Table

In [31]:
```
df_241UDS0096["City"] = df_241UDS0096["City"].replace(["Hyderabad"],
["Hyd"])
df_241UDS0096[df_241UDS0096["City"] == "Hyd"].head(2)
```

Out[31]: Columns 1–8 of 16

	Movie_Title	Genre	Language	Release_Year	Budget_Cr	Tickets_Sold	Discount_Percent
Movie_ID							
M1001	Pushpa 2: The Rule	Action	Telugu	2024	180	650	5.0
M1009	Ala Vaikunthapurramuloo	Drama	Telugu	2020	100	1450	11.0

Columns 9–15 of 16 · first column repeated

	IMDb_Rating	Available_Seats	Payment_Method	City	Ticket_Price	Total_Box_Office_241UDS0096	Discount_Amount_241UDS0096
Movie_ID							
M1001	4.2	18	UPI	Hyd	250	162500	8125.0
M1009	7.3	16	UPI	Hyd	250	362500	39875.0

Columns 16–16 of 16 · first column repeated

	Net_Revenue_241UDS0096
Movie_ID	
M1001	154375.0
M1009	322625.0

Step 25 — Updating a Specific Movie Value

In [32]:
```
# Update the ticket price for movie M1001
df_241UDS0096.loc["M1001", "Ticket_Price"] = 300
print(df_241UDS0096.loc["M1001", "Ticket_Price"])
```

```
300
```

Step 26 — Changing Column Names

In [33]:
```
df_241UDS0096 = df_241UDS0096.rename(columns={
    "Movie_Title": "Film_Name",
    "City": "Location",
    "IMDb_Rating": "Rating",
    "Net_Revenue_241UDS0096": "Revenue"
})
df_241UDS0096.head(1)
```

Out[33]:

Columns 1–8 of 16

	Film_Name	Genre	Language	Release_Year	Budget_Cr	Tickets_Sold	Discount_Percent
Movie_ID							
M1001	Pushpa 2: The Rule	Action	Telugu	2024	180	650	5.0

Columns 9–15 of 16 · first column repeated

	Rating	Available_Seats	Payment_Method	Location	Ticket_Price	Total_Box_Office_241UDS0096	Discount_Amount_241UDS0096
Movie_ID							
M1001	4.2	18	UPI	Hyd	300	162500	8125.0

Columns 16–16 of 16 · first column repeated

	Revenue
Movie_ID	
M1001	154375.0

Step 27 — Final DataFrame

In [34]:
```
df_241UDS0096.head()
```

Out[34]:

Columns 1–8 of 16

Movie_ID	Film_Name	Genre	Language	Release_Year	Budget_Cr	Tickets_Sold	Discount_Percent
M1001	Pushpa 2: The Rule	Action	Telugu	2024	180	650	5.0
M1002	Kalki 2898 AD	Sci-Fi	Telugu	2024	600	1050	10.0
M1003	Devara: Part 1	Action	Telugu	2024	300	520	8.0
M1004	Hanu-Man	Fantasy	Telugu	2024	40	390	12.0
M1005	Salaar	Action	Telugu	2023	270	1250	7.0

Columns 9–15 of 16 · first column repeated

Movie_ID	Rating	Available_Seats	Payment_Method	Location	Ticket_Price	Total_Box_Office_241UDS0096	Discount_Amount_241UDS0096
M1001	4.2	18	UPI	Hyd	300	162500	8125.0
M1002	8.5	35	Card	Vijayawada	250	262500	26250.0
M1003	6.8	22	UPI	Visakhapatnam	250	130000	10400.0
M1004	7.8	41	Cash	Pune	250	97500	11700.0
M1005	6.5	14	Card	Mumbai	250	312500	21875.0

Columns 16–16 of 16 · first column repeated

Movie_ID	Revenue
M1001	154375.0
M1002	236250.0
M1003	119600.0
M1004	85800.0
M1005	290625.0

Step 28 — Final Dataset Information

In [35]:
```
df_241UDS0096.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 30 entries, M1001 to M1030
Data columns (total 15 columns):
 #   Column                        Non-Null Count  Dtype  
---  ------                        --------------  -----  
 0   Film_Name                     30 non-null     object 
 1   Genre                         30 non-null     object 
 2   Language                      30 non-null     object 
 3   Release_Year                  30 non-null     int64  
 4   Budget_Cr                     30 non-null     int64  
 5   Tickets_Sold                  30 non-null     int64  
 6   Discount_Percent              30 non-null     float64
 7   Rating                        30 non-null     float64
 8   Available_Seats               30 non-null     int64  
 9   Payment_Method                30 non-null     object 
 10  Location                      30 non-null     object 
 11  Ticket_Price                  30 non-null     int64  
 12  Total_Box_Office_241UDS0096   30 non-null     int64  
 13  Discount_Amount_241UDS0096    30 non-null     float64
 14  Revenue                       30 non-null     float64
dtypes: float64(4), int64(6), object(5)
memory usage: 4.8+ KB
```

RESULT

By using the Pandas library, data manipulation operations—including selecting rows and columns, creating calculated columns, filtering, sorting, updating values, and renaming columns—were implemented on a movie dataset.